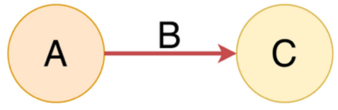
Research Progress

Relation-Level Structural Analysis of Predictive Multiplicity in Knowledge Graph Embedding for Link Prediction

知識グラフ埋め込みにおける予測多重性をもたらす
関係レベル構造要因の解明

M2+ 趙 怡名

Background: Knowledge graph embedding link prediction

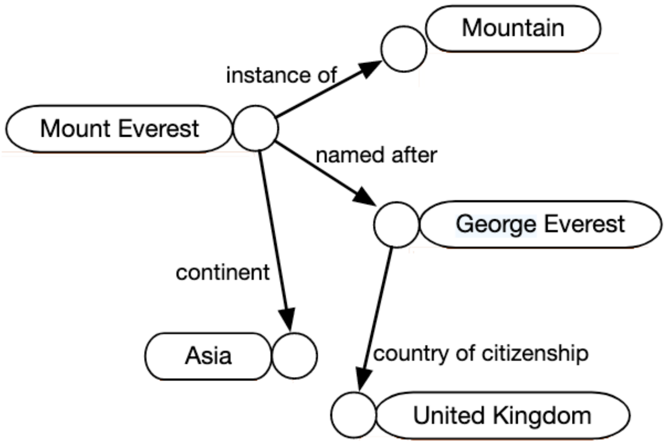


Triple

A: head entity

B: relation

C: tail entity



Example of Knowledge graph[1]

Knowledge graph link prediction

(Tail prediction & Head prediction)

Goal: $(h, r, ?)$ or $(?, r, t)$

(Paris, capital_of , ?)

Two possible ranking outcomes

Ranking	Entity
1	**France**
2	Germany
3	Italy
…	…
10	Spain
11	China
…	…

$Hit@10$

Ranking	Entity
1	Germany
2	Spain
3	Italy
…	…
10	China
11	**France**
…	…

$Miss@10$

Evaluation metric: $Hits@10$

The proportion of test queries for which this top-10 success condition is satisfied.

Predictive multiplicity[2]

Even when models have similar overall performance, they may still disagree on whether a specific query is a top-K hit or miss.

This phenomenon is quantified by **ambiguity** and **discrepancy**.

Toy Example: Same Overall Performance with Query-Level Disagreement

Query q	Model A	Model B	Model C
q_1: `(Paris, capital_of, ?)`	Hit	Hit	Miss
q_2: `(CompanyA, has_supplier, ?)`	Hit	Miss	Hit
q_3: `(PersonB, born_in, ?)`	Miss	Hit	Hit
q_4: `(FilmC, directed_by, ?)`	Hit	Hit	Hit
$Hits@10$	75%	75%	75%

Ambiguity (α)

Measures how many queries are unstable across similarly performing models.

(Among all test queries, how many receive at least one conflicting decision?)

Here: $\alpha = \frac{3}{4} = 0.75$

Discrepancy (δ)

Measures the worst-case disagreement rate between the baseline and one competing model.

(For one competing model, how large can the overall disagreement rate be?)

Here: Model B vs. A: $\frac{2}{4}$ Model C vs. A : $\frac{2}{4}$

$$\delta = \max\left(\frac{2}{4}, \frac{2}{4}\right) = 0.5$$

Key idea of multiplicity

Similar overall performance does not imply identical query-level decisions.

Research Focus: From Prior Work to Thesis Question

Original paper

- Formalizes predictive multiplicity in KGE link prediction
- Introduces ambiguity and discrepancy as evaluation metrics
- Shows that multiplicity exists empirically, and voting can mitigate it

My thesis question

- The original paper establishes the existence of predictive multiplicity in KGE link prediction.
- My thesis extends this line of inquiry by asking whether predictive multiplicity is systematically associated with relation-level structure.
- Accordingly, the focus shifts from a global phenomenon to relation-level analysis.
- **Main question: Which relation-level factors are associated with stronger predictive multiplicity?**

Method: Thesis Setting and Relation-Level Analysis Pipeline

Thesis setting

- Dataset: FB15k-237
- Models: RotatE[3] and TransE[4]
- Repeated runs of the same model family
- Analysis unit: relation-level rather than only dataset-level/global-level

FB15k-237: a standard KGE benchmark with less inverse-relation leakage than FB15k, making it more suitable for structural analysis of link prediction.

RotatE / TransE: two representative KGE models used here as a focused cross-model comparison under the current thesis scope.

Pipeline

- Run repeated training with different random seeds
- Compute query-level top-10 hit / miss outcomes
- Aggregate them into relation-level metrics
- Attach relation-level factors
- Compare which factors are associated with stronger multiplicity

Selection principle

Predictive multiplicity is a top-K ranking instability.

I therefore focus on relation-level factors that may affect answer-space ambiguity, structural support, or data sparsity.

1. Mapping property

How many distinct heads/tails are typically associated with a relation.

Typical categories

$1-1$, $1-N$, $M-1$, $M-N$ (one-to-one, one-to-many, many-to-one, many-to-many)

Intuition

Changes the number of plausible candidates on the prediction side.
When the prediction side is the many-side, the top-K boundary may become less stable.

Example

- `has_capital`: closer to 1-1
- `has_child`: closer to 1-N
- `born_in`: closer to M-1
- `language_spoken_in`: closer to M-N

2. Inverse

Whether a relation has a clear reverse partner.

Intuition

Reverse-side evidence may constrain prediction behavior and make repeated runs more consistent.

Example

- `parent_of` ⇔ `child_of`
- `capital_of` ⇔ `has_capital`

3. Symmetry

Whether (h, r, t) is often supported by (t, r, h) under the same relation.

Intuition

Internal bidirectional evidence may provide consistent signals for the same entity pair and affect ranking stability.

Example

- `married_to`
- `adjacent_to`

4. Relation frequency (support/ sparsity factor)

How often a relation appears in the training graph.

Control question

Is multiplicity caused by relation structure, or simply by data sparsity?

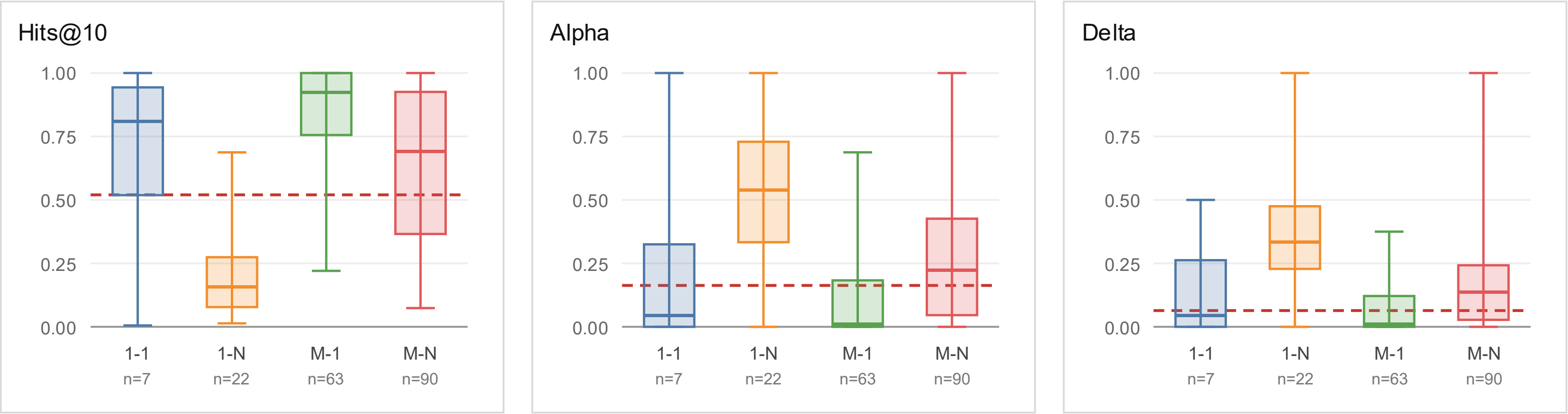
Role

Support / sparsity control.

Result 1: Mapping Property Strongly Affects Multiplicity

RotatE tail-side

Query: $(h, r, ?)$



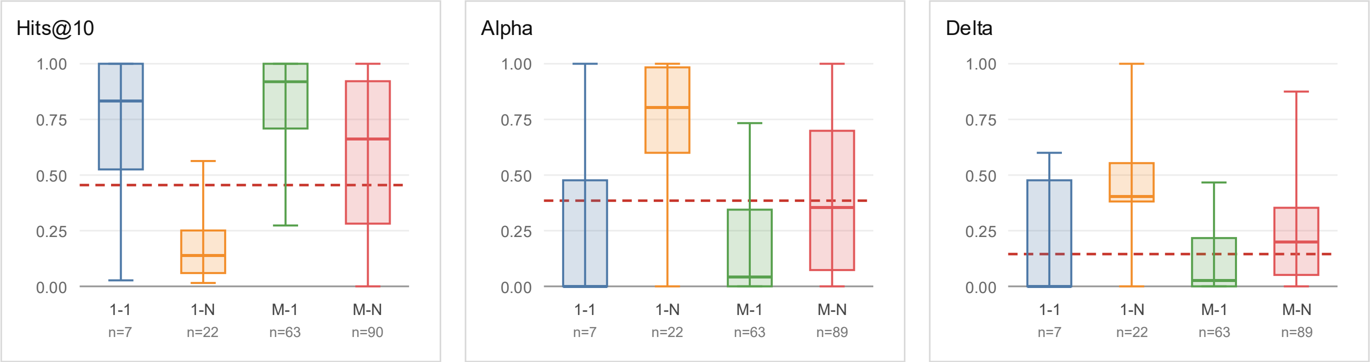
The red dashed line shows the corresponding result from the original paper.

Key finding: Multiplicity follows the answer-space structure of the relation.

- 1-N and M-1 form opposite tail-side regimes:
  1-N has a less constrained tail answer space, while M-1 has a more constrained tail target.
- 1-N therefore shows the clearest instability:
  low $Hits@10$ with high α / δ
- M-N is also tail-many, but less extreme:
  its intermediate behavior may reflect both larger candidate space and broader structural redundancy.

Result 1: Mapping Property Strongly Affects Multiplicity

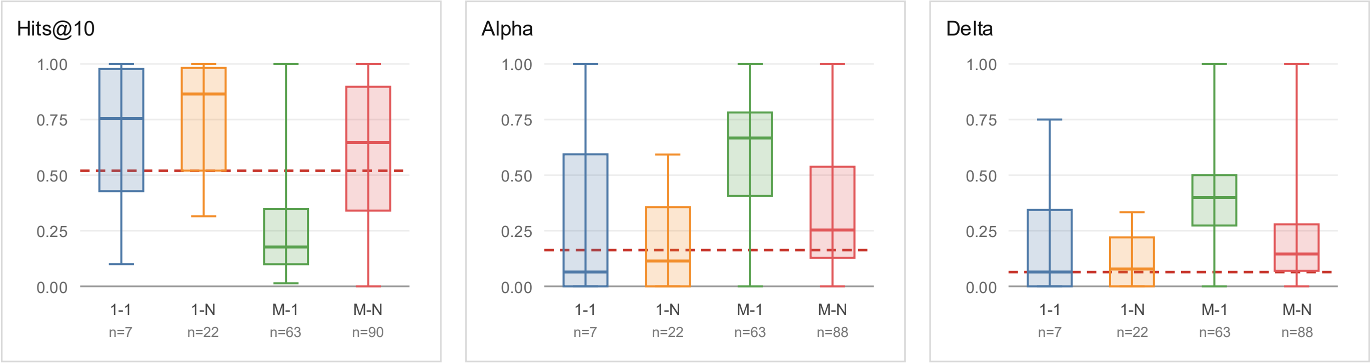
Cross-model check: TransE tail-side   Query: $(h, r, ?)$



The red dashed line shows the corresponding result from the original paper.

- The same tail-side contrast remains visible in TransE.
- M-1 remains the most stable regime, while 1-N remains the most multiplicity-prone regime.
- The mapping-property effect is not specific to a single model (RotatE).

Prediction-side check: RotatE head-side   Query: $(?, r, t)$



The red dashed line shows the corresponding result from the original paper.

- The pattern reverses when switching from tail prediction to head prediction.
- 1-N becomes stable regime, while M-1 becomes the most multiplicity-prone regime.
- Multiplicity follows the predicted side, not a fixed mapping-type label.

Result 2: High-Confidence Inverse-Like Relations Are More Stable

Key finding

No clean global trend, but high-confidence inverse-like relations form a stable group.

Inverse-Like Support and Predictive Stability

Model	Relations	n_rel	test triples	Hits@10	Alpha	Delta
RotatE	All relations (global baseline)	182	20262	0.5797	0.1630	0.0640
RotatE	High mutual inverse-like strength (> 0.5)	10	494	0.6838	0.0971	0.0627
RotatE	High inverse-like clarity (> 0.5)	5	61	0.8000	0.0000	0.0000
TransE	All relations (global baseline)	182	20262	0.5585	0.3850	0.1450
TransE	High mutual inverse-like strength (> 0.5)	10	494	0.6866	0.1361	0.0813
TransE	High inverse-like clarity (> 0.5)	5	61	0.8000	0.0000	0.0000

`mutual_inverse_strength` : bidirectional reverse-support strength.
`inverse_clarity`: confidence that the best reverse partner is clearly distinguishable.

- No clean global monotonic trend is observed.
- High-confidence inverse-like subgroups show higher $Hits@10$ and lower α / δ than the global baseline.
- The cleanest subgroup is defined by inverse_clarity > 0.5, where both RotatE and TransE show $\alpha = 0$ and $\delta = 0$.
- A clear reverse partner provides a strong reverse-side structural constraint. The same entity pair can be supported from two directions: (h, r, t) and (t, r_{inv}, h) , making top-K rankings less sensitive to random training variation.

Example

`/education/educational_institution_campus/educational_institution` → institution of the campus

`/education/educational_institution/campuses` → campuses of the institution

mutual_inverse_strength	=	0.8026
inverse_clarity (two directions)	=	0.7770/0.7710
hits@10	=	1.0000
alpha	=	0.0000
delta	=	0.0000

Result 3: Symmetry Alone Does Not Define a Stable Regime

Global check: weak Spearman correlations

Global Spearman Correlation

Model	n	Spearman(Hits)	Spearman(Alpha)	Spearman(Delta)
RotatE	182	0.0122	0.1310	0.1292
TransE	182	0.0117	0.0425	0.0520

Bucket check: high-symmetry group is not more stable

Bucket Comparison

Model	Group	n	Hits@10	Alpha	Delta
RotatE	Zero symmetry	164	0.5815	0.1283	0.0636
RotatE	Weak nonzero symmetry (0, 0.5]	6	0.5047	0.2089	0.1129
RotatE	High symmetry (> 0.5)	12	0.5924	0.1618	0.0848
TransE	Zero symmetry	164	0.5605	0.3307	0.1838
TransE	Weak nonzero symmetry (0, 0.5]	6	0.4725	0.4436	0.2736
TransE	High symmetry (> 0.5)	12	0.5742	0.3512	0.2119

- **Global association is weak**:
  Spearman correlations are close to zero for both models.
- **Bucket comparison is non-monotonic**:
  weak nonzero symmetry shows lower $Hits@10$ and higher multiplicity.
- **High symmetry does not form a stable regime**:
  $Hits@10$ recovers, but α and δ remain relatively high.
- **Interpretation**:
  symmetry may impose type-level constraints, but not pair-specific reverse support.

Summary

- Predictive multiplicity shows relation-level structural heterogeneity.
- Mapping property provides the strongest and most directional signal:
  multiplicity follows the predicted side of the relation.
- High-confidence inverse-like pairs form a locally stable group,
  but inverse-like support is not a clean global factor.
- Symmetry alone does not define a stable regime.
- Overall, predictive multiplicity should be analyzed not only as a global model-level phenomenon,
  but also through relation-level structural factors.

Future work

- **Relation frequency as a control factor**
  Examine whether the mapping-property effect remains after controlling for relation support and sparsity.
- **Broader validation**
  Extend the analysis to more datasets and additional KGE model families.
- **From diagnosis to mitigation**
  Use relation-level instability signals to design targeted voting or adaptive ensemble strategies.

Reference

[1] Lv, Yana, and Haomiao Bao. “NSE-KGC: A Knowledge Graph Completion Method Based on Neighborhood Semantic Evidence.” In *2024 5th International Conference on Big Data & Artificial Intelligence & Software Engineering (ICBASE)*, 287–90, 2024.

[2] Zhu Y, Potyka N, Nayyeri M, et al. Predictive multiplicity of knowledge graph embeddings in link prediction[C]. *Findings of the Association for Computational Linguistics: EMNLP 2024*. 2024: 334-354.

[3] Sun Z, Deng Z H, Nie J Y, et al. Rotate: Knowledge graph embedding by relational rotation in complex space[J]. arXiv preprint arXiv: 1902. 10197, 2019.

[4] Bordes A, Usunier N, Garcia-Duran A, et al. Translating embeddings for modeling multi-relational data[J]. Advances in neural information processing systems, 2013, 26.

Mapping Property

Relation edge set

$$E_r = \{(h,t) \mid (h,r,t) \in \mathcal{G}_{\text{train}}\}$$

Average tail connectivity:

$$\text{avgTail}(r) = \frac{1}{|\{\, h \mid \exists t,\ (h,r,t) \in \mathcal{G}_{\text{train}}\}|} \sum_{h} |\{\, t \mid (h,r,t) \in \mathcal{G}_{\text{train}}\}|$$

Average head connectivity:

$$\text{avgHead}(r) = \frac{1}{|\{\, t \mid \exists h,\ (h,r,t) \in \mathcal{G}_{\text{train}}\}|} \sum_{t} |\{\, h \mid (h,r,t) \in \mathcal{G}_{\text{train}}\}|$$

Classification rule:

$1-1$:	$avgTail(r) \leq 1.5$	and	$avgHead(r) \leq 1.5$
$1-N$:	$avgTail(r) > 1.5$	and	$avgHead(r) \leq 1.5$
$M-1$:	$avgTail(r) \leq 1.5$	and	$avgHead(r) > 1.5$
$M-N$:	$avgTail(r) > 1.5$	and	$avgHead(r) > 1.5$

Symmetry

Edge set:

$$E_r = \{(h,t) \mid (h,r,t) \in \mathcal{G}_{\text{train}}\}$$

Symmetric-supported edge set:

$$E_r^{\text{sym}} = \{(h,t) \in E_r \mid (t,h) \in E_r\}$$

Raw symmetry score:

$$\text{Sym}(r) = \frac{|E_r^{\text{sym}}|}{|E_r|}$$

To remove self-loop inflation, the main metric excludes edges with h = t.

$$E_r^{\text{sym},\neg\text{loop}} = \{(h,t) \in E_r \mid h \neq t,\ (t,h) \in E_r\}$$

Excluding-self edge set:

$$E_r^{\neg\text{loop}} = \{(h,t) \in E_r \mid h \neq t\}$$

Main symmetry score:

$$\text{Sym}^{\neg\text{loop}}(r) = \frac{|E_r^{\text{sym},\neg\text{loop}}|}{|E_r^{\neg\text{loop}}|}$$

Inverse

For each relation r, let E_r be its directed edge set in the training graph.
The reversed edge set E_r^{rev} is a flipped version of E_r, and does not require (t, r, h) to exist in the graph.

$$E_r = \{(h,t) \mid (h,r,t) \in \mathcal{G}_{\text{train}}\}$$

$$E_r^{\text{rev}} = \{(t,h) \mid (h,r,t) \in \mathcal{G}_{\text{train}}\}$$

Baseline directional inverse score:
This measures how much of relation r_1 is supported by the reversed edges of relation r_2.

$$s(r_1 \to r_2) = \frac{|E_{r_1} \cap E_{r_2}^{\text{rev}}|}{|E_{r_1}|}$$

Mutual inverse score:

$$s_{12} = s(r_1 \to r_2), \qquad s_{21} = s(r_2 \to r_1)$$

$$s_{\text{mut}}(r_1, r_2) = \begin{cases} \frac{2 s_{12} s_{21}}{s_{12}+s_{21}}, & s_{12} + s_{21} > 0 \\ 0, & \text{otherwise} \end{cases}$$

Mutual inverse strength:
For each relation, we keep the strongest mutual inverse-like partner.

$$\text{MutInvStrength}(r) = \max_{r' \in \mathcal{R},\, r' \neq r} s_{\text{mut}}(r, r')$$

Inverse clarity:
This measures whether the best inverse-like partner is clearly better than the second-best candidate.

$$\text{InvClarity}(r) = \max\Big(0,\ \text{MutInvStrength}(r) - \text{SecondBestMutInvStrength}(r)\Big)$$